माधव प्रौद्योगिकी एवं विज्ञान संस्थान, ग्वालियर (म.प्र.), भारत
MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR, (M.P.), INDIA
Deemed to be University
(Declared under Distinct Category by Ministry of Education, Government of India)
NAAC ACCREDITED WITH A++ GRADE

Department of Mathematics and Computing

SEMESTER PROFICIENCY

Jan– June 2026, MAC second year

By- Arpit Yadav

Enrollment No. - BTMC24O1032

Data Science

This course offers practical exposure to handling real-world data, from collecting and cleaning raw datasets to extracting meaningful insights through visualization and statistical analysis. It also builds the ability to develop and evaluate predictive models, helping in solving real-life problems using data-driven approaches.

- Introduction to Data Science and it’s scope
- Data Science lifestyle
- Types of data: Structured, Semi-structured, Unstructured
- Data formats: TXT, CSV, JSON, XLSX

- How raw data is converted into meaningful insights
- Difference between various data types and formats
- Basics of reading and handling data files
- Importance of data in decision-making

Applications:

- Netflix analyzes user viewing data (CSV/JSON) to recommend shows
- Healthcare systems store patient records in structured databases
- Social media platforms handle unstructured data like images and text

Database Management System

This course offers a strong foundation in organizing, storing, and managing large-scale data efficiently. It enhances skills in writing optimized SQL queries, designing normalized databases, and understanding how modern systems handle transactions, concurrency, and distributed data.

- Database approach vs traditional file system
- Database users and DBA
- Data models: Hierarchical, Network, Relational
- Entity-Relationship (ER) model

- Why databases are more efficient than file systems
- Role of DBMS in data consistency and security
- How ER diagrams represent real-world entities
- Identification of entities, attributes, and relationships

Applications:

- Banking systems use relational databases for transactions
- College management systems use ER models for students and faculty
- E-commerce platforms manage products, users, and orders using DBMS

Software Engineering

This course offers a structured approach to developing software systems by focusing on requirement analysis, design methodologies, and development models. It also strengthens understanding of project planning, risk management, and testing strategies to ensure high-quality and maintainable software.

- Introduction to Software Engineering
- Software Development Life Cycle (SDLC)
- Models: Waterfall, Spiral, RAD, Prototyping

- Structured approach to software development
- Selection of SDLC models based on project type
- Importance of planning, risk management, and user involvement

Applications:

- Banking software uses Waterfall for stability
- Startups use Agile/RAD for rapid development
- Mobile apps often follow iterative models

Transform and Vector Calculus

This course offers insight into advanced mathematical tools used to analyze and solve engineering problems. It develops the ability to apply transforms and vector operations in areas like signal processing, differential equations, and physical system analysis.

- Introduction to Fourier Series
- Dirichlet’s conditions
- Even and odd functions
- Half-range Fourier series

- How periodic functions are expressed as trignometric series
- Decomposition of signals into sine and cosine components
- Applications of Fourier series in engineering

Applications:

- Signal processing in mobile communication
- Sound and image compression
- Electrical engineering for analyzing waveforms

Number Theory and Cryptography

This course offers a blend of mathematical theory and its application in secure communication. It builds understanding of number theoretic concepts and their use in cryptographic algorithms, enabling secure data transmission and modern encryption techniques.

- Divisibility theory
- Modular arithmetic
- Prime numbers and congruences
- Residue systems

- How periodic functions are expressed as trignometric series
- Decomposition of signals into sine and cosine components
- Applications of Fourier series in engineering

Applications:

- RSA encryption in secure web communication
- ATM PIN verification systems
- Digital signatures for authentication

THANK YOU!!

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- Sound and image compression
- Electrical engineering for analyzing waveforms

Number Theory and Cryptography - Unit 1

- Divisibility theory
- Modular arithmetic
- Prime numbers and congruences
- Residue systems

- Mathematical foundation behind encryption
- Importance of prime numbers in security
- Use of modular arithmetic in cryptography

- RSA encryption in secure web communication
- ATM PIN verification systems
- Digital signatures for authentication

ECDSA and Schnorr Signatures in Cryptography

“Secure digital signatures based on elliptic curve mathematics”

ECDSA (Elliptic Curve Digital Signature Algorithm) and **Schnorr Signatures** are cryptographic techniques used to verify authenticity and integrity of digital messages.
They provide secure authentication with smaller key sizes and high efficiency.

Key Characteristics:

- Based on Elliptic Curve Cryptography (ECC)
- High security with low computational cost
- Used for digital signatures and authentication

ECDSA

- Widely used standard
- Non-deterministic signatures
- Used in Bitcoin and TLS

Schnorr Signatures

- Simpler and more secure design
- Deterministic signatures
- Supports signature aggregation

Applications

- Blockchain and cryptocurrencies
- Secure communication protocols
- Digital certificates and authentication

Green Databases in DBMS

“Efficient data management with minimal environmental impact”

Green Databases focus on reducing energy consumption and carbon footprint while maintaining database performance. They aim to make data storage and processing more sustainable, reducing power usage, carbon footprint and resource wastage without comprimising database performance.

Core Concepts

- Energy-efficient query processing
- Reduced storage redundancy
- Optimized resource utilization

Key Techniques

- Query optimization- executes queries using minimal CPU and memory
- Data compression- reduces storage space and disk usage.
- Virtualization and cloud efficiency- shares resources efficiently among multiple users

Benefits

- Lower power consumption
- Reduced operational cost
- Environment-friendly computing

Applications

- Large-scale data centers
- Cloud-based database systems
- Sustainable IT infrastructure

Data Pre-Processing in Data Science

“Better data leads to better models”

Data pre-processing is the process of preparing raw data for analysis and machine learning. Real-world data is often incomplete, noisy, inconsistent and unstructured. Data pre-processing improves data quality and further enhances accuracy and produce reliable results.

Major Steps

- Data cleaning
- Data integration
- Data transformation
- Data reduction

Common Tasks

- Handling missing values
- Removing duplicates
- Normalization and scaling
- Encoding categorical data

Why It Is Important

- Reduces noise and inconsistency
- Improves algorithm performance
- Ensures reliable results

Applications

- Machine learning pipelines
- Data analytics
- Predictive modeling

Coupling and Cohesion in Software Engineering

“Good design has low coupling and high cohesion.”

Coupling and **Cohesion** are key principles used to measure software design quality. They help in building maintainable and scalable systems.

Coupling

- Degree of dependency between modules
- Low coupling is preferred
- Types:

1. Tight coupling- modules are highly dependent on each other
2. Loose coupling- modules interact with minimal dependency (preferred)

Cohesion

- Degree to which elements of a module belong together
- High cohesion is preferred
- Types:

1. Functional- all elements contribute to a single, well defined task
2. Sequential- output of one function is input to another function
3. Logical- similar kinds of operations are grouped together

Benefits

- Easier maintenance
- Better readability
- Improved reusability

Applications

- Object-oriented design
- Modular programming
- Large-scale software systems

Fourier Series in Image and Video Processing

“Transforming signals from time domain to frequency domain.”

Fourier Series decomposes signals into sine and cosine components for analysis.
It is widely used in image and video processing.

Key Concepts

- Frequency domain representation
- Periodic signal decomposition
- Spectral analysis

Uses in Image Processing

- Image compression
- Noise removal
- Image enhancement

Uses in Video Processing

- Motion analysis
- Video compression
- Frame filtering

Applications

- JPEG and MPEG compression
- Medical imaging
- Computer vision systems

THANK YOU!!